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1: //With numbers
2:
3: #include <stdio.h>
4:
5: void moveBubbleUp(void);
6: void moveBubbleDown(void);
7: void printArray(void);
8: void generateArray(void);
9: void numsToLetters(void);
10:
11: int arr[12];
12: const size = 12;
13:
14: int main() {
15:     generateArray();
16:     printf("Array before sorting: \n");
17:     printArray();
18:     bubbleSort();
19:     return 0;
20: }
21:
22: void bubbleSort(void){
23:     moveBubbleUp();
24:     moveBubbleDown();
25: }
26:
27: void moveBubbleUp(void) {
28:     for (int step = 0; step < 11; ++step) {
29:         for (int i = 0; i < 11 - step; ++i) {
30:             if (arr[i] > arr[i + 1]) {
31:                 int temp = arr[i];
32:                 arr[i] = arr[i + 1];
33:                 arr[i + 1] = temp;
34:             }
35:         }
36:     }
37:     printf("Sorted array in ascending order: \n");
38:     printArray();
39: }
40:
41: void moveBubbleDown(void){
42:     for (int step = 0; step < 11; step++) {
43:         for (int i = 0; i < 11 - step; i++) {
44:             if (arr[i] < arr[i + 1]) {
45:                 int temp = arr[i];
46:                 arr[i] = arr[i + 1];
47:                 arr[i + 1] = temp;
48:             }
49:         }
50:     }
51:     printf("Sorted array in descending order: \n");
52:     printArray();
53: }
54:
55: void generateArray(void){
56:     for(int i=0; i<12; i++){
57:         int num = rand();
58:         arr[i]=num;
59:     }
60: }

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61:
62: void printArray(void) {
63:     for (int i = 0; i < size; ++i) {
64:         printf("%d ", arr[i]);
65:     }
66:     printf("\n");
67: }
68:
69: //With Letters
70: #include <stdio.h>
71:
72: void bubbleSort(void);
73: void moveBubbleUp(void);
74: void moveBubbleDown(void);
75: void printArray(void);
76: void generateArray(void);
77:
78: int arr[12];
79: const size = 12;
80:
81: int main() {
82:     generateArray();
83:     printf("Array before sorting: \nStatus: Unsorted\n");
84:     printArray();
85:     bubbleSort();
86:     return 0;
87: }
88:
89: void bubbleSort(void){
90:     moveBubbleUp();
91:     printf("\nStatus: Sorted");
92:     moveBubbleDown();
93:     printf("\nStatus: Sorted");
94: }
95:
96: void moveBubbleUp(void) {
97:     for (int step = 0; step < 11; ++step) {
98:         for (int i = 0; i < 11 - step; ++i) {
99:             if (arr[i] > arr[i + 1]) {
100:                 int temp = arr[i];
101:                 arr[i] = arr[i + 1];
102:                 arr[i + 1] = temp;
103:             }
104:         }
105:     }
106:     printf("Sorted array in ascending order: \n");
107:     printArray();
108: }
109:
110: void moveBubbleDown(void){
111:     for (int step = 0; step < 11; step++) {
112:         for (int i = 0; i < 11 - step; i++) {
113:             if (arr[i] < arr[i + 1]) {
114:                 int temp = arr[i];
115:                 arr[i] = arr[i + 1];
116:                 arr[i + 1] = temp;
117:             }
118:         }
119:     }
120:     printf("Sorted array in descending order: \n");

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121:     printArray();
122: }
123:
124: void generateArray(void){
125:     for(int i=0; i<12; i++){
126:         arr[i]=rand()% (90 - 65 + 1) + 65;
127:     }
128: }
129:
130: void printArray(void) {
131:     for (int i = 0; i < size; ++i) {
132:         printf("%c ", arr[i]);
133:     }
134:     printf("\n");
135: }
136:
137:
```